

Results Summary

Brief note: training results for one saved iSFS logistic run on the real dataset.

Run Setup

Item	Value
Dataset	real
Outer max iter	10
Block A max iter	40
Alpha/Beta subproblem max iter	25 / 25
Learning rate alpha	0.01
Learning rate beta	0.01
Lambda beta	0.01
Alpha l1 radius	1.0
Decision threshold	0.5
Initial cost vector	0.5,0.5
Baseline cost vector	0.5,0.5
Final learned cost vector	[0.7650, 0.9240]

Overall Performance

Metric	Value
Samples	500
Valid predictions	500
TP	179
TN	190

Metric	Value
FP	75
FN	56
Accuracy	0.7380
Precision	0.7047
Recall / Sensitivity	0.7617
Specificity	0.7170
F1 score	0.7321
Balanced accuracy	0.7393
Negative predictive value	0.7724
False positive rate	0.2830
False negative rate	0.2383
MCC	0.4779
Brier score	0.1755
Log loss	0.5244
Probability range	0.0161 to 0.9878

BCD Summary

Recorded iterations	Accepted cost steps	Final cost vector
4	3	[0.7650, 0.9240]

Iter	Accepted	Pred. increase	Actual increase	Ratio	Radius	Current objective	Trial objective
1	True	0.0801	0.1315	1.6418	0.0200	0.3018	0.4333
2	True	0.1403	0.0412	0.2935	0.0400	0.4333	0.4745
3	True	0.0115	0.0107	0.9310	0.0400	0.4745	0.4852
4	False	0.0000	0.0000	-inf	0.0800	0.4852	0.4852

Learned Alpha Weights

Profile	source1	source2	source3
source1	0.1937		
source1, source2	0.1466	0.6848	
source1, source2, source3	0.0641	0.0000	0.9359
source1, source3	0.0752		0.9248
source2		0.8908	
source2, source3		0.0263	0.9737
source3			1.0000

Beta Summary

Source	Features	L1 norm	L2 norm	Nonzero coeffs
source1	10	0.4716	0.2755	7
source2	8	0.2890	0.1620	4
source3	25	2.0169	0.6417	14

Exact Profile Results

Code	Samples	Accuracy	Precision	Recall	Specificity	F1	BCE	Cost-BCE
001	27	0.7407	0.6667	0.7273	0.7500	0.6957	0.5149	0.4263
010	27	0.6667	0.6250	0.7692	0.5714	0.6897	0.6685	0.5622
011	61	0.7705	0.8065	0.7576	0.7857	0.7812	0.5001	0.4290
100	41	0.6585	0.6087	0.7368	0.5909	0.6667	0.6013	0.5004
101	56	0.7143	0.6786	0.7308	0.7000	0.7037	0.5375	0.4492
110	75	0.6800	0.6098	0.7576	0.6190	0.6757	0.5904	0.4896

Code	Samples	Accuracy	Precision	Recall	Specificity	F1	BCE	Cost-BCE
111	213	0.7793	0.7573	0.7800	0.7788	0.7685	0.4728	0.3966

Optimization Group Results

Code	Samples	Accuracy	Precision	Recall	Specificity	F1	BCE	Cost-BCE
001	357	0.7703	0.7558	0.7647	0.7754	0.7602	0.4955	0.4171
010	376	0.6356	0.6105	0.6480	0.6244	0.6287	0.6436	0.5399
011	274	0.7774	0.7727	0.7669	0.7872	0.7698	0.4825	0.4073
100	385	0.6390	0.5882	0.7303	0.5604	0.6516	0.6336	0.5271
101	269	0.7695	0.7462	0.7698	0.7692	0.7578	0.4862	0.4074
110	288	0.6840	0.6280	0.7744	0.6065	0.6936	0.6092	0.5069
111	213	0.7793	0.7573	0.7800	0.7788	0.7685	0.4728	0.3966